

Meric Kavlicoglu

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EDUCATION

Purdue University

Class of May 2027

Bachelor of Science in Mechanical Engineering

West Lafayette, IN

- Pursuing Four Year BS/MS
- **Dean's List:** 3.77 GPA
- Relevant Coursework: Mechanics of Materials, Dynamics, Graphical Communication and Spatial Analysis, Partial Differential Equations, Machine Design, Controls

TECHNICAL SKILLS

Software: Siemens NX, Fusion 360 CAM/CAD, Ansys Workbench Finite Element Analysis (FEA), Python, C++, Simulink

Manufacturing and Prototyping: Manual/CNC Milling and Turning, Geometric Dimensioning and Tolerancing, 3D Printing

EXPERIENCE

Clickbond Inc.

May 2026 - August 2026

Product Development Intern

Carson City, NV

- Developed tooling for assembly of parts to **Lockheed Standard**
- Performed **Monte Carlo Statistical Analysis** to analyze tolerance stackups and reduce failure modes from tooling, optimizing performance from a **2 σ to 5 σ process**
- Programmed a **4-axis Fanuc Robot** arm with palletization for assembly
- Developed a **PID Controller** for a linear actuator with current feedback for precise **Force Control within 1 lb**

Advanced Materials and Devices

May 2025 - August 2025

Mechanical Engineering Intern

Reno, NV

- Designed and optimized a multilateration algorithm for **high-speed data processing (2000 kHz)**, reducing runtime by **1000x** and improving output accuracy
- Developed a **3-D non-monotonic model** to analyze previously unmodeled DoD data, enabling deeper insight into **Blast Wave and Mach Stem behavior**, improving data interpretation workflows
- Built a custom test rig and developed a 7-day test plan **increasing test efficiency by 50%**, defining procedures, instrumentation, and data-collection intervals to ensure consistent, high-fidelity performance measurements

ME 164: Geometric and Annotation Modeling TA

August 2025 - Current

Undergraduate Teaching Assistant

West Lafayette, IN

- Teaching Siemens NX to 36 students, providing individualized design feedback and guiding them through CAD modeling best practices
- Demonstrated practical applications of **Geometric Dimensioning and Tolerancing** and instructed students on ASME and ISO engineering drawing standards to improve manufacturability and communication

PROJECTS

Purdue Space Program Liquids | Siemens NX, Fusion 360 CAM, Python, GitHub

October 2025 - Present

- **System Owner** on pneumatic actuator used on the Pathfinder rocket's oxidizer and fuel valves, integration with other rocket subsystems, emphasis on **design for manufacturing**
- **Manufacturing Owner** for fuel injector, programming CAM for CNC milling high precision parts, satisfying tight design tolerances of **1 thousandth**, measuring basic flatness and cylindricity GD&T
- Leading a team of 3 engineers, coordinating actuator sizing calculations, system integration, and alignment with the main rocket assembly

Go-Kart Analysis and Design | Fusion 360 CAD/CAM, Python, DFM/DFA

January 2026 - Present

- Defined system requirements for **custom Go-Kart**
- Developed a prototype model with **DFM analysis** for manufacturing without CNC Machining
- Analyzed vehicle dynamics and powertrain to **optimize design for system requirements**